

MTHOBISI NXUMALO

SOFTWARE ENGINEER | FULL STACK DEVELOPER

Durban, KwaZulu-Natal, South Africa | 071 927 9462 | mthobisinx30@gmail.com
portfolio.rephinasoftware.com | github.com/mthobisi30 | linkedin.com/in/mthobisi30

PROFESSIONAL PROFILE

Software Engineer and Full Stack Developer experienced in designing, building and operating production web, mobile and desktop systems across manufacturing, industrial inspection, e-commerce and financial services. Own the full SDLC from requirements and architecture through database design, development, CI/CD, deployment and production support. Strong in TypeScript, React, Next.js, Node.js and PostgreSQL, with additional experience in Flutter, React Native, Python, C#.NET, Docker, AWS and Microsoft Azure.

CORE TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart, C#, Kotlin, C++, SQL

Frontend & Mobile: React, Next.js, Flutter, React Native, Capacitor, Tailwind CSS, shadcn/ui, HTML5, CSS3

Backend & APIs: Node.js, Flask, ASP.NET Core, REST API design, MVC architecture

Data: PostgreSQL, Microsoft SQL Server, Oracle, SQLite, Prisma ORM, SQLAlchemy, schema design, query optimisation and migrations

DevOps & Cloud: Docker, Coolify, GitHub Actions, CI/CD, Git, AWS and Microsoft Azure

Security & Practices: JWT, OAuth, RBAC, tenant isolation, audit logging, full SDLC, system architecture, technical documentation and Agile/Scrum

PROFESSIONAL EXPERIENCE

Wesant Manufacturing / Vision Elevators - Full Stack Developer | Pinetown, KZN | Sep 2025 - Present

- Own the architecture, development, deployment and ongoing support of internal and customer-facing web and mobile applications, taking business requirements through to production delivery.
- Build production applications using Next.js, React and TypeScript, alongside cross-platform Flutter/Dart and native Android/Kotlin solutions where required.
- Administer PostgreSQL and Microsoft SQL Server, including schema design, query optimisation, indexing, migrations, data integrity and automated backups.
- Deploy and operate containerised applications with Docker on self-hosted Coolify infrastructure, maintaining CI/CD pipelines, releases, uptime and production monitoring.

Rephina Software - Founder & Full Stack Developer | Durban, KZN | 2023 - Present

- Deliver custom web, mobile and desktop applications, dashboards, APIs and workflow automation for commercial clients, with responsibility spanning discovery, architecture, implementation and production support.
- Delivered complex multi-surface platforms including MPIA Services, an industrial NDT inspection and operations platform used across administrative, client and field workflows.

Hamtern Financial Services - Learner, IT & Systems Support | Bryanston, Gauteng | Jan 2024 - Dec 2024

- Built workflow automation using Microsoft Power Automate, Power Apps and Power BI to reduce manual administration across departments.

EDUCATION & CERTIFICATIONS

BSc Information Technology - North-West University, Potchefstroom | 2020 - 2023

Short-Term Insurance (NQF Level 4) - Apex Academy | 2024

National Senior Certificate (Matric) - King Edward VII School, Johannesburg | 2015 - 2019

Certifications: Diploma in Cyber Security - Alison (2024); freeCodeCamp: Responsive Web Design and JavaScript Algorithms & Data Structures (2023)

SELECTED ENGINEERING PROJECTS

MPIA Services - Industrial NDT Inspection & Operations Platform

Next.js 15 | TypeScript | Flutter | Tauri 2 | PostgreSQL | Prisma | NextAuth | SQLite

Architected and delivered a multi-surface industrial inspection platform that centralises administrative, client and field operations around one PostgreSQL data model. The solution spans seven application surfaces: admin web and desktop applications, a client portal across web, desktop and mobile, a dedicated field mobile application, and a public-facing marketing site.

- Designed the shared system architecture and central PostgreSQL database so jobs, inspection records, reports, invoices, users and client data remain consistent across web, desktop and mobile applications.
- Engineered an offline-first field workflow in Flutter using local SQLite storage, enabling technicians to capture inspection data without connectivity and automatically synchronise records when a connection is restored.
- Built tenant-scoped client access and granular role-based permissions so organisations and users can access only the jobs, reports, invoices and inspection information relevant to them.
- Implemented audit logging across operational records to improve traceability of changes and user activity throughout inspection and administrative workflows.
- Delivered Tauri 2 desktop applications with Ed25519-signed installers and automatic update support, extending the platform beyond browser and mobile environments while maintaining a unified back-end architecture.
- Owned the platform end to end through Rephina Software, including requirements analysis, solution architecture, database design, application development, deployment and ongoing technical support.

ElevateTech - Manufacturing PIM, Supplier Portal, E-commerce & Mobile Platform

Next.js | React | React Native | TypeScript | PostgreSQL | Prisma | Docker | Coolify | Paystack

Designed and developed an integrated product and commerce ecosystem for an elevator-components business, connecting product information management, supplier workflows, customer commerce and mobile access to a shared product database. The platform replaces fragmented product processes with connected systems for maintaining catalogue data, supplier information, inventory and customer-facing product content.

- Delivered four connected production surfaces: a Product Information Management (PIM) system, supplier portal, customer e-commerce store and React Native mobile application, all operating against a shared PostgreSQL product data model.
- Built version-controlled product information and change-history audit trails, providing traceability as product specifications, catalogue content and related data are updated over time.
- Implemented bulk CSV import and export workflows to support high-volume catalogue maintenance and reduce manual product-data administration.
- Developed role-based dashboards and access controls for different operational users, with workflows covering product catalogue management, supplier interactions and business administration.
- Implemented automated inventory synchronisation and connected catalogue/SKU workflows so product availability and customer-facing information can remain aligned across platform surfaces.
- Supported e-commerce account, order and payment workflows, including approved business-customer access and payment processing integration, as part of the broader commerce platform.
- Containerised and deployed services with Docker on self-hosted Coolify infrastructure, using automated database migrations and release workflows designed to minimise production downtime.